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Theopractical Sheet 2

1.Question 1

Your task is to gather information on the memory and storage components used in one of the following devices.

• If you own this device, you could try to unscrew the shielding and take some pictures of the components
on the circuit to extract the information.

• If you do not own this device, you find the necessary information in the internet.

Devices to choose from:

1. AVM Fritz!Box 7490 (no FCC ID)
2. Google Nest Hub Max (FCC ID: A4R-H2A)
3. Amazon Echo 1st Generation (FCC ID: ZWJ-0823)
4. Samsung SmartThings Hub (FCC ID: R3Y-STH-ETH201)

My Selection:Amazon Echo 1st Generation (FCC ID: ZWJ-0823)

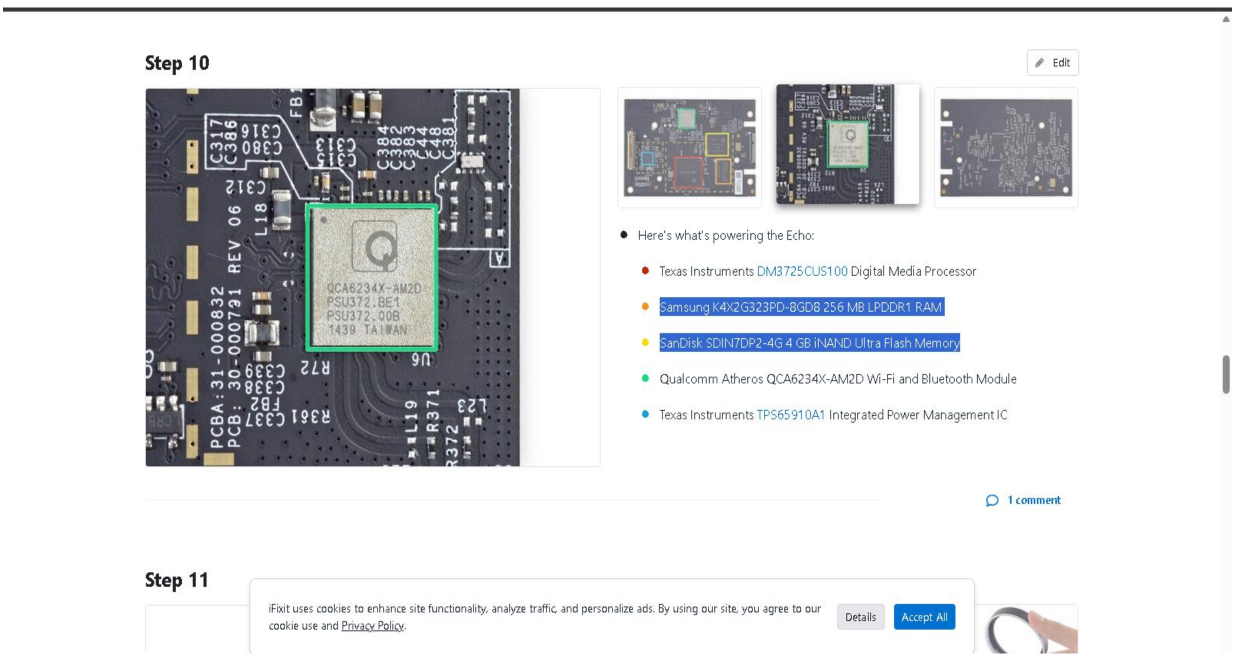
a) What type of memory/storage components are soldered on any of the circuits of the product? Select all that apply. Ignore very small components that are a part of other components (e.g., the DRAM buffer of a SSD) and would not occur separately in a part list of the product.

Ans:eMMC Module,DDR-1 SDRAM

Since I don’t have the Amazon Echo 1st Generation device I searched for teardowns for the device and https://www.ifixit.com/Teardown/Amazon+Echo+Teardown/33953 helped me out in this teardown it was shown that

Samsung K4X2G323PD-8GD8: This is a 256 MB LPDDR1 RAM module, which is a type of volatile memory used for temporary data storage during device operation.

SanDisk SDIN7DP2-4G: This is a 4 GB iNAND Ultra Flash Memory module which is technially a eMMC module, a type of non-volatile storage used to store the device's firmware and user data.



b)b) Give detail information (product number, memory/disk type, capacity) about all memory/storage components that you identified on any of the circuits of the product.

• For components that are there more than once, it suffices to include it once.

• If there are several variants of the product and the FCC ID does not suffice to specify which variant is meant, decide on one.

• Ignore very small components that are a part of other components (e.g., the DRAM buffer of a SSD) and would not occur separately in a part list of the product.

• Include the volatility class you would assign this component to as a string.

• Include the analysis type of digital forensics that is most suitable for analyzing this component as a string.

• Include the data classes according to the classification by semantics that you expect to be stored on this component as an array of strings.

A product number may contain digits, letters, dashes and whitespaces. Do not include any vendor names. For all "<choose from below>" fields, make sure to choose a correct type from the provided lists as strings. For the capacity, make sure to add the unit.

Your array should have, depending on your choice, between 2 (incl.) and 4 (incl.) entries. You get a full point for every correct entry and half a point if at least three of the fields match per array element. The points are doubled if necessary, depending on your choice.

Ans:[
```
 {
   "product-number": "K4X2G323PD-8GD8",
   "type": "DDR1-SDRAM",
   "capacity": "256 MiB",
   "volatility": "Volatile (broad)",
   "forensic-analysis-type": "Live Analysis",
   "data": ["Primary data", "Program data"]
 },
 {
   "product-number": "SDIN7DP2-4G",
   "type": "eMMC",
   "capacity": "4 GiB",
   "volatility": "Persistent",
   "forensic-analysis-type": "Postmortem Analysis",
   "data": ["Primary data", "Configuration Data", "Log Data"]
 }
]
```

Samsung K4X2G323PD-8GD8 is volatile as data is lost when power is off and SanDisk SDIN7DP2-4G is Persistant which basically means retains data without power.

2.Question 2

In the materials section of CMS, you find an encrypted ZIP archive and a DOCX document. Crack the passwords of both files. Give the password to open the files and the flags that are inside the files. Include the tool you have used and the command that you executed to obtain the password in your proof-of-work file. For flags, include the identifier "FLAG" and the braces.

Ans:a) What is the password of the ZIP archive?
guessme

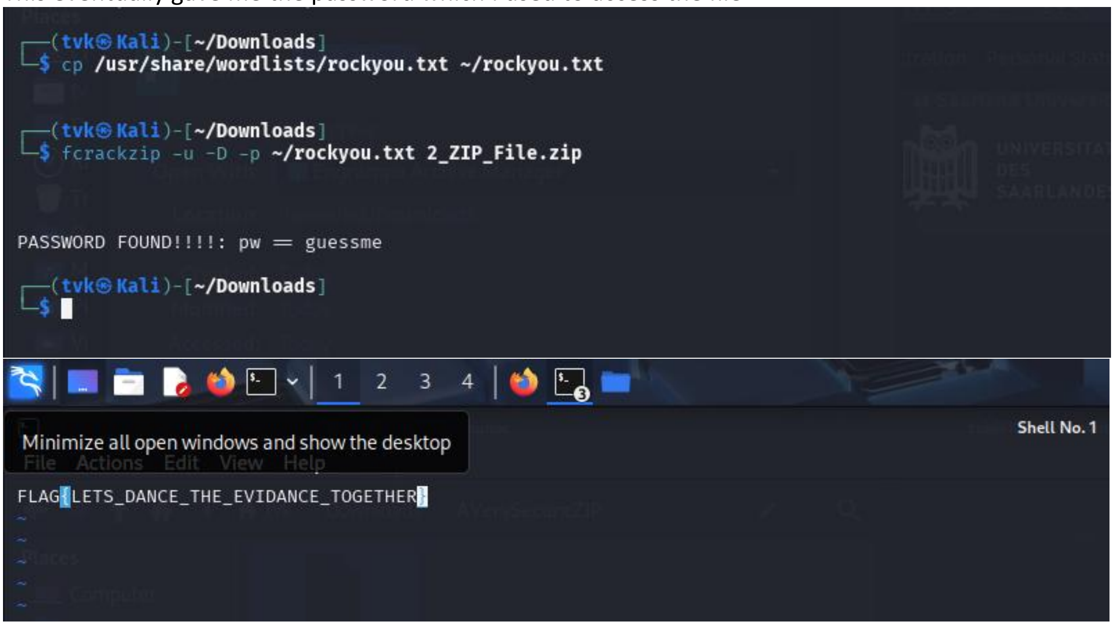
b)FLAG{LETS_DANCE_THE_EVIDANCE_TOGETHER}

The tools I used were fcrackzip for zip file

I used a password list rockyou.txt and decompress it using :
sudo gunzip /usr/share/wordlists/rockyou.txt.gz

Then after copying the file to the folder I used fcrackzip to crack the zip file

fcrackzip -u -D -p ~/rockyou.txt 2_ZIP_File.zip
This eventually gave me the password which I used to access the file

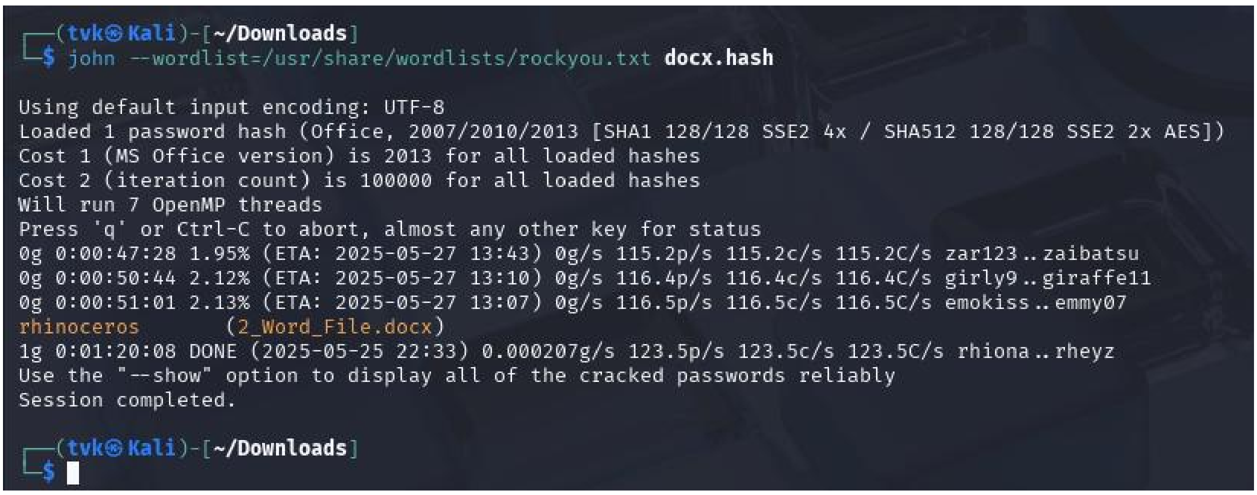


c)What is the password of the DOCX document?
Ans:rhinoceros

d)What is the flag inside the DOCX document?
Ans:FLAG{I_AM_OUT_OF_WORD}

The tools I used was John the ripper to crack the password for docx file.
Since I already have the wordlist from above I directly used John the ripper to crack the password.
john --wordlist=/usr/share/wordlists/rockyou.txt docx.hash

This eventually gave me the password which I used to access the file



3. Select all correct statements.

Ans:While Flash Cells are horizontally stacked in 2D NAND Flash, they are vertically stacked in 3D NAND Flash. (This is true since that is the difference between 2D NAND and 3D NAND)

The more bits a Flash Cell can store, the less Program/Erase cycles it endures.(More accurate voltage levels are needed as the number of bits per cell increases.The cell can be written or erased fewer times before wearing out because these levels become less stable.

Hence More bits per cell= less Durability)

4.What does "collision resistance" mean? Select the correct option. Notation: "value" means preimage.

Ans:It is practically infeasible to find two values with the same hash

5.Assume you have a raw, uncompressed image of a hard drive that you can load to Autopsy for analysis. Which steps of the investigative process model as shown in chapter 4 can be executed inside Autopsy? Assume that you are using the default installation without any additional plugins or extensions.

Ans:Preservation (Autpsy doesn't change the original instead this it works on copies of data which are basically forensic images.

Recovery (Autopsy can be used to locate hidden data, partitions and erased files)

Harvesting(Autopsy can be used to extract important information from documents,logs etc)

Reduction (Autopsy can also remove unneccsary data)

Organization and search (Autopsy can also search and sort for files)

Analysis (Metadata can be examined)

Reporting(Autopsy can generate investigation reports after the analysis)

The Investigative Process Model